



CHE 109: Engineering Chemistry – I
Summer-2025
Basics of Biochemistry
Chapter 10

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19.14 Amino Acids

Chapter 29 discusses the synthesis of amino acids and their conversion to proteins.

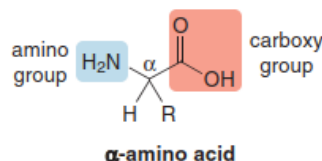


Humans can synthesize only 10 of the 20 amino acids needed for protein synthesis. The remaining 10, called **essential amino acids**, must be obtained from the diet and consumed on a regular, almost daily basis. Vegetarian diets must be carefully balanced to obtain all the essential amino acids. Grains—wheat, rice, and corn—are low in lysine, and legumes—beans, peas, and peanuts—are low in methionine, but a combination of these foods provides all the needed amino acids. Thus, a diet of corn tortillas and beans, or rice and tofu, provides all essential amino acids. A peanut butter sandwich on wheat bread does, too.

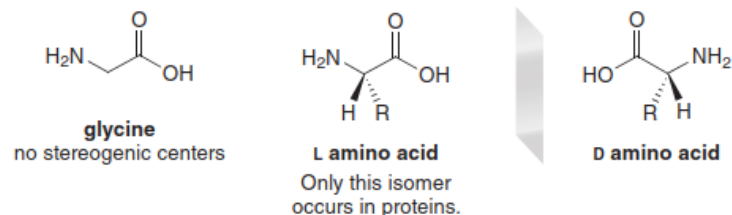
Amino acids, one of four kinds of small biomolecules that have important biological functions in the cell (Section 3.9), also undergo proton transfer reactions.

19.14A Introduction

Amino acids contain two functional groups—an amino group (NH_2) and a carboxy group (COOH). In most naturally occurring amino acids, the amino group is bonded to the α carbon, and so they are called **α -amino acids**. Amino acids are the building blocks of proteins, biomolecules that comprise muscle, hair, fingernails, and many other biological tissues.



The 20 amino acids that occur naturally in proteins differ in the identity of the R group bonded to the α carbon. **The simplest amino acid, called glycine, has $\text{R} = \text{H}$.** When the R group is any other substituent, **the α carbon is a stereogenic center**, and there are two possible enantiomers.



Amino acids exist in nature as only one of these enantiomers. Except when the R group is CH_2SH , the stereogenic center on the α carbon has the *S* configuration. An older system of nomenclature names the **naturally occurring enantiomer of an amino acid as the L isomer**, and its **unnatural enantiomer the D isomer**.

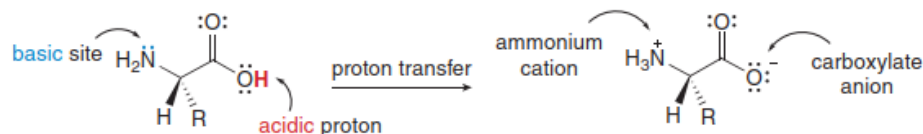
Figure 29.2 The 20 naturally occurring amino acids

Neutral amino acids					
Name	Structure	Abbreviations	Name	Structure	Abbreviations
Alanine		Ala A	Phenylalanine*		Phe F
Asparagine		Asn N	Proline		Pro P
Cysteine		Cys C	Serine		Ser S
Glutamine		Gln Q	Threonine*		Thr T
Glycine		Gly G	Tryptophan*		Trp W
Isoleucine*		Ile I	Tyrosine		Tyr Y
Leucine*		Leu L	Valine*		Val V
Methionine*		Met M			
Acidic amino acids			Basic amino acids		
Name	Structure	Abbreviations	Name	Structure	Abbreviations
Aspartic acid		Asp D	Arginine*		Arg R
Glutamic acid		Glu E	Histidine*		His H
			Lysine*		Lys K

Essential amino acids are labeled with an asterisk (*).

29.1B Acid–Base Behavior

Recall from Section 19.14B that an amino acid has both an acidic and a basic functional group, so proton transfer forms a salt called a **zwitterion**.



This neutral form of an amino acid does **not** really exist.

The zwitterion is neutral.

This salt is the neutral form of an amino acid.

This form exists at $\text{pH} \approx 6$.

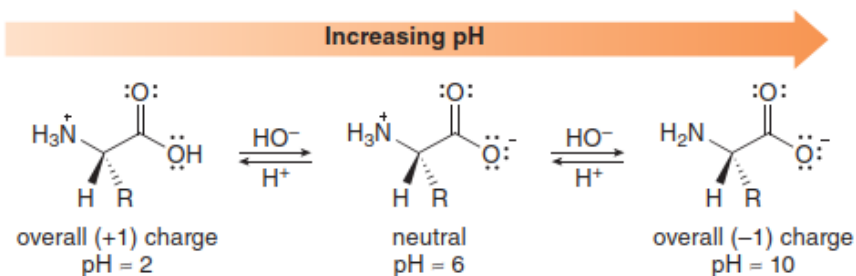
The structures in Figure 29.2 show the charged form of the amino acids at the physiological pH of the blood.

- Amino acids do not exist to any appreciable extent as uncharged neutral compounds. They exist as salts, giving them high melting points and making them water soluble.

Amino acids exist in different charged forms, as shown in Figure 29.3, depending on the pH of the aqueous solution in which they are dissolved. For neutral amino acids, the overall charge is +1, 0, or -1. Only at $\text{pH} \sim 6$ does the zwitterionic form exist.

Figure 29.3

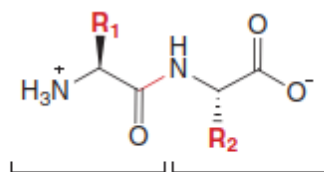
How the charge of a neutral amino acid depends on the pH



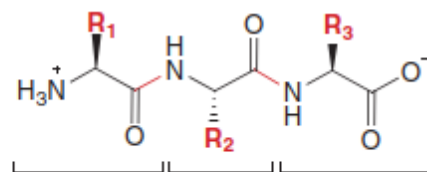
29.5 Peptides

When amino acids are joined together by amide bonds, they form larger molecules called **peptides** and **proteins**.

- A *dipeptide* has two amino acids joined together by *one* amide bond.
- A *tripeptide* has three amino acids joined together by *two* amide bonds.



Two amino acids joined together.



Three amino acids joined together.

[Amide bonds are drawn in red.]

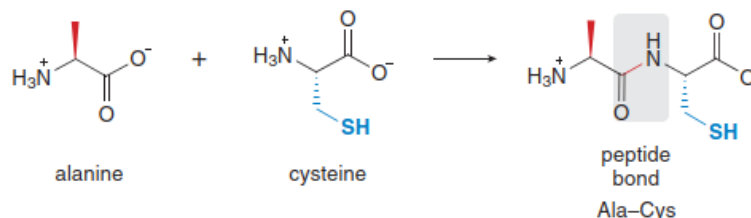
Polypeptides and **proteins** both have many amino acids joined together in long linear chains, but the term **protein** is usually reserved for polymers of more than 40 amino acids.

- The amide bonds in peptides and proteins are called *peptide bonds*.
- The individual amino acids are called *amino acid residues*.

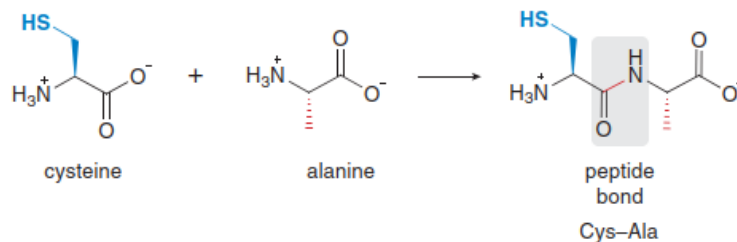
29.5A Simple Peptides

To form a dipeptide, the amino group of one amino acid forms an amide bond with the carboxy group of another amino acid. Because each amino acid has both an amino group and a carboxy group, **two different dipeptides can be formed**. This is illustrated with alanine and cysteine.

[1] The COO^- group of alanine can combine with the NH_3^+ group of cysteine.



[2] The COO^- group of cysteine can combine with the NH_3^+ group of alanine.



These compounds are **constitutional isomers** of each other. Both have a free amino group (protonated as NH_3^+) at one end of their chains and a free carboxy group (deprotonated as a carboxylate anion, COO^-) at the other.

- The amino acid with the free amino group is called the *N-terminal amino acid*.
- The amino acid with the free carboxy group is called the *C-terminal amino acid*.

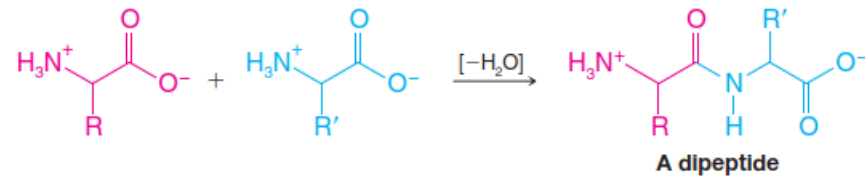
By convention, **the N-terminal amino acid is always written at the left end of the chain and the C-terminal amino acid at the right**. The peptide can be abbreviated by writing the one- or three-letter symbols for the amino acids in the chain from the N-terminal to the C-terminal end. Thus, Ala-Cys has alanine at the N-terminal end and cysteine at the C-terminal end, whereas Cys-Ala has cysteine at the N-terminal end and alanine at the C-terminal end. Sample Problem 29.1 shows how this convention applies to a tripeptide.

24.4 POLYPEPTIDES AND PROTEINS

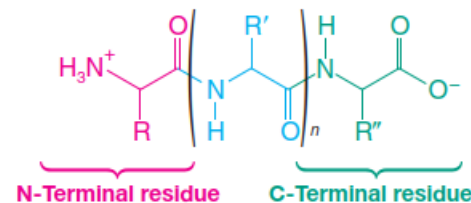
Amino acids are polymerized in living systems by enzymes that form amide linkages from the amino group of one amino acid to the carboxyl group of another.

- A molecule formed by joining amino acids together is called a **peptide**, and the amide linkages in them are called **peptide bonds** or **peptide linkages**. Each amino acid in the peptide is called an **amino acid residue**.

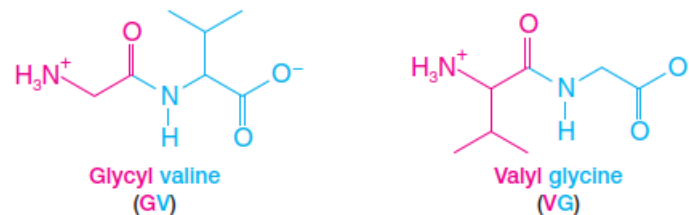
Peptides that contain 2, 3, a few (3–10), or many amino acids are called **dipeptides**, **tripeptides**, **oligopeptides**, and **polypeptides**, respectively. **Proteins** are polypeptides consisting of one or more polypeptide chains.



Polypeptides are **linear polymers**. One end of a polypeptide chain terminates in an amino acid residue that has a free $-NH_3^+$ group; the other terminates in an amino acid residue with a free $-CO_2^-$ group. These two groups are called the **N-terminal** and the **C-terminal residues**, respectively:



- By convention, we write peptide and protein structures with the N-terminal amino acid residue on the left and the C-terminal residue on the right:



24.3 Proteins

Proteins are *biological polymers of small molecules called amino acids*. Their molecular weights range from about 6000 amu to 250,000 amu, so proteins can be very large molecules. Many proteins also contain non-amino-acid components such as metal ions (for example, Fe^{2+} , Zn^{2+} , Cu^{+} , and Mg^{2+}) or certain complex organic molecules usually derived from vitamins. (Vitamins are organic molecules necessary in very small quantities for normal cell structure and metabolism.)

Proteins are very important molecules in cells and organisms, playing both structural and functional roles. The proteins of bone and connective tissue, for example, are of major structural importance. Some proteins are enzymes, catalyzing specific biochemical reactions; some transport materials in the blood stream or across biological membranes; and some (hormones) carry chemical messages to coordinate the body's activities. *Insulin* and *glucagon*, for example, are protein hormones made in the pancreas and secreted to regulate the body's blood-sugar level.

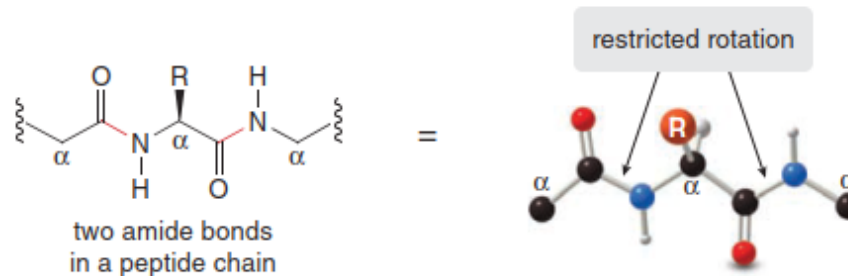
29.9 Protein Structure

Now that you have learned some of the chemistry of amino acids, it's time to study proteins, the large polymers of amino acids that are responsible for so much of the structure and function of all living cells. We begin with a discussion of the **primary, secondary, tertiary, and quaternary structure** of proteins.

29.9A Primary Structure

The *primary structure* of proteins is the particular sequence of amino acids that is joined together by **peptide bonds**. The most important element of this primary structure is the **amide bond**.

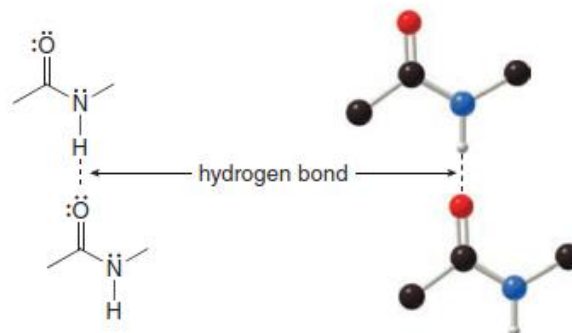
- Rotation around the amide C–N bond is *restricted* because of electron delocalization, and the *s-trans* conformation is the more stable arrangement.
- In each peptide bond, the N–H and C=O bonds are directed 180° from each other.



Although rotation about the amide bonds is restricted, **rotation about the other σ bonds in the protein backbone is not**. As a result, the peptide chain can twist and bend into a variety of different arrangements that constitute the secondary structure of the protein.

29.9B Secondary Structure

The three-dimensional conformations of localized regions of a protein are called its **secondary structure**. These regions arise due to hydrogen bonding between the N–H proton of one amide and C=O oxygen of another. Two arrangements that are particularly stable are called the **α -helix** and the **β -pleated sheet**.



α -Helix

The **α -helix** forms when a peptide chain twists into a right-handed or clockwise spiral, as shown in Figure 29.8. Four important features of the α -helix are as follows:

- [1] **Each turn of the helix has 3.6 amino acids.**
- [2] **The N–H and C=O bonds point along the axis of the helix.** All C=O bonds point in one direction, and all N–H bonds point in the opposite direction.
- [3] **The C=O group of one amino acid is hydrogen bonded to an N–H group four amino acid residues farther along the chain.** Thus, hydrogen bonding occurs between two amino acids *in the same chain*. Note, too, that the hydrogen bonds are parallel to the axis of the helix.
- [4] **The R groups of the amino acids extend outward** from the core of the helix.

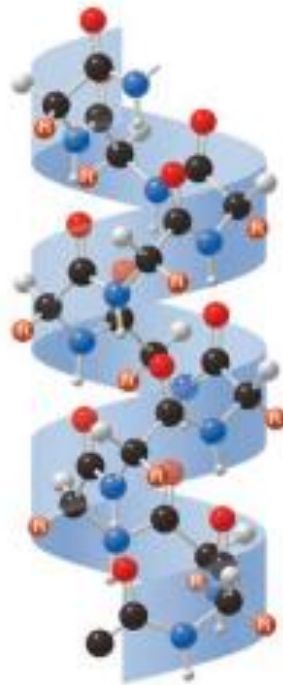
An α -helix can form only if there is rotation about the bonds at the α carbon of the amide carbonyl group, and not all amino acids can do this. For example, proline, the amino acid whose nitrogen atom forms part of a five-membered ring, is more rigid than other amino acids, and its C_α –N bond cannot rotate the necessary amount. Additionally, it has no N–H proton with which to form an intramolecular hydrogen bond to stabilize the helix. Thus, proline cannot be part of an α -helix.

Both the myosin in muscle and α -keratin in hair are proteins composed almost entirely of α -helices.

Figure 29.8

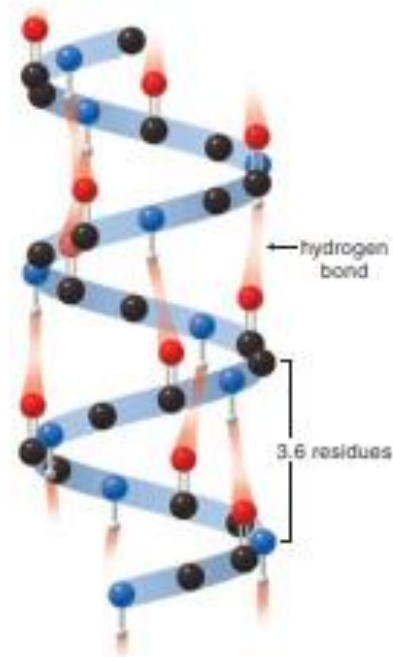
Two different illustrations
of the α -helix

a. The right-handed α -helix



All atoms of the α -helix are drawn in this representation. All C=O bonds are pointing up and all N-H bonds are pointing down.

b. The backbone of the α -helix



Only the peptide backbone is drawn in this representation. The hydrogen bonds between the C=O and N-H of amino acids four residues away from each other are shown.

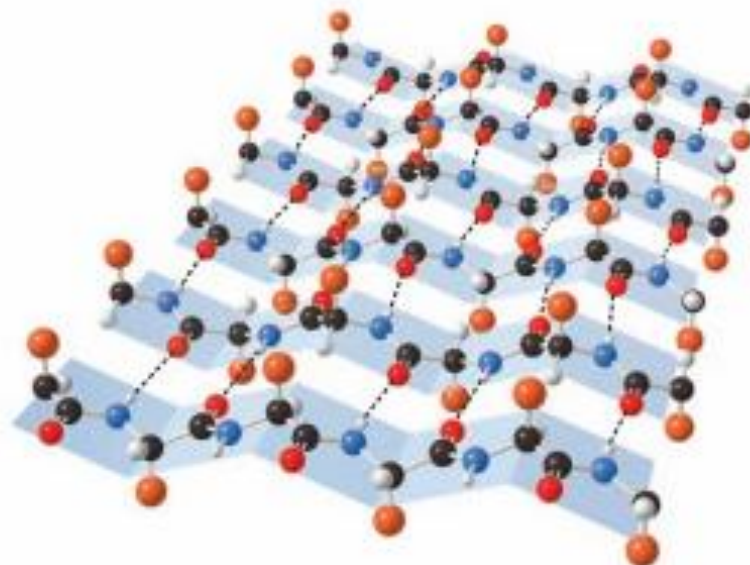
β -Pleated Sheet

The β -pleated sheet secondary structure forms when two or more peptide chains, called **strands**, line up side-by-side, as shown in Figure 29.9. All β -pleated sheets have the following characteristics:

- [1] The C=O and N-H bonds lie in the plane of the sheet.
- [2] Hydrogen bonding often occurs between the N-H and C=O groups of nearby amino acid residues.
- [3] The R groups are oriented above and below the plane of the sheet, and alternate from one side to the other along a given strand.

Figure 29.9

Three-dimensional structure of the β -pleated sheet



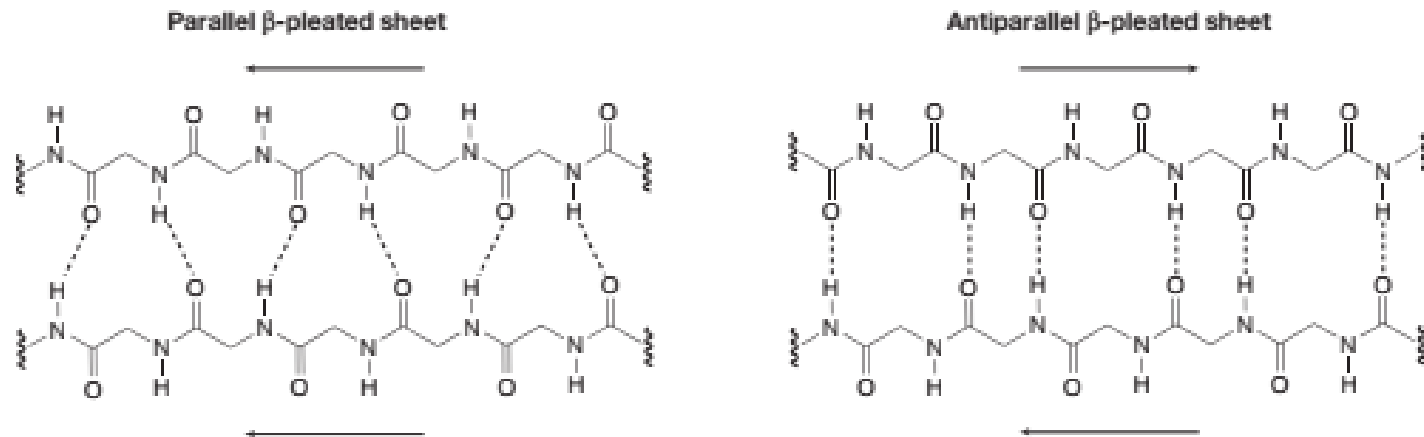
- The β -pleated sheet consists of extended strands of the peptide chains held together by hydrogen bonding. The C=O and N-H bonds lie in the plane of the sheet, and the R groups (shown as orange balls) alternate above and below the plane.

The β -pleated sheet arrangement most commonly occurs with amino acids with small R groups, like alanine and glycine. With larger R groups steric interactions prevent the chains from getting close together and so the sheet cannot be stabilized by hydrogen bonding.

The peptide strands of β -pleated sheets can actually be oriented in two different ways, as shown in Figure 29.10.

- In a *parallel* β -pleated sheet, the strands run in the same direction from the N- to C-terminal amino acid.
- In an *antiparallel* β -pleated sheet, the strands run in the *opposite* direction.

Figure 29.10 The parallel and antiparallel forms of the β -pleated sheet



The two peptide chains are arranged in the same direction. Hydrogen bonds occur between N-H and C=O bonds in adjacent chains.

The two peptide chains are arranged in opposite directions. Hydrogen bonding between the N-H and C=O groups still holds the two chains together.

[Note: R groups on the carbon chain are omitted for clarity.]

28.1 Introduction

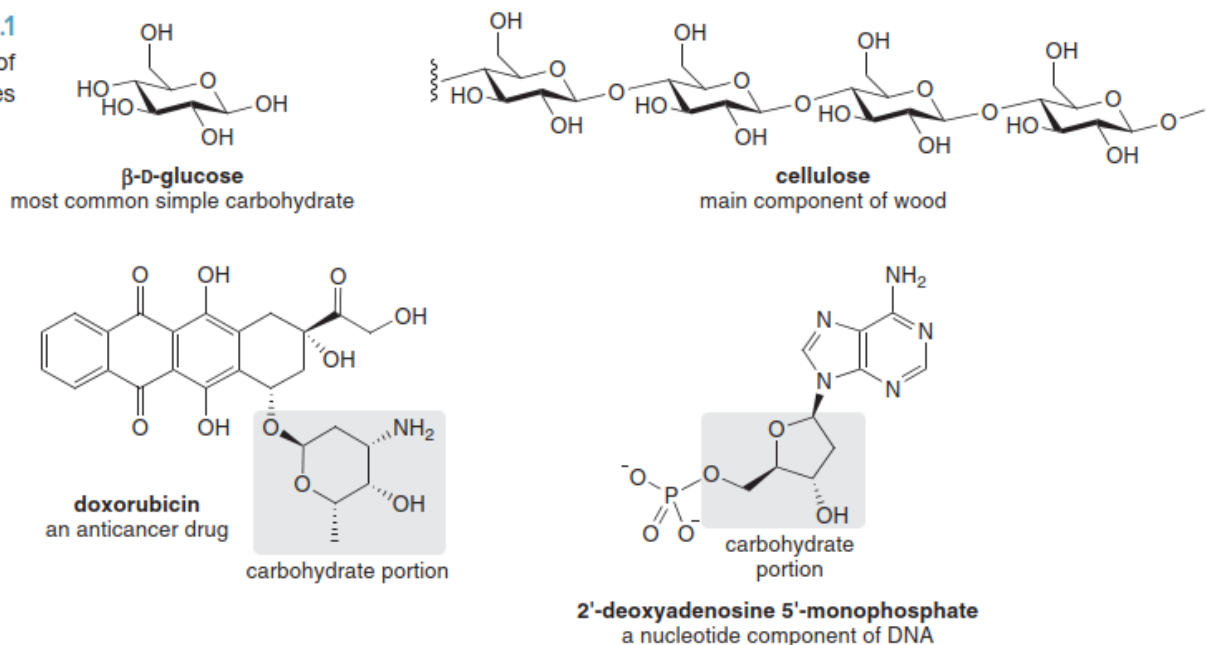
Carbohydrates were given their name because molecular formulas of simple carbohydrates could be written as $C_n(H_2O)_n$, making them **hydrates of carbon**.

Carbohydrates such as glucose and cellulose were discussed in Sections 5.1, 6.4, and 21.17.

Carbohydrates, commonly referred to as sugars and starches, are polyhydroxy aldehydes and ketones, or compounds that can be hydrolyzed to them. The cellulose in plant stems and tree trunks and the chitin in the exoskeletons of arthropods and mollusks are both complex carbohydrates. Four examples are shown in Figure 28.1. They include not only glucose and cellulose, but also doxorubicin (an anticancer drug) and 2'-deoxyadenosine 5'-monophosphate (a nucleotide base from DNA), both of which have a carbohydrate moiety as part of a larger molecule.

Carbohydrates are storehouses of chemical energy. They are synthesized in green plants and algae by **photosynthesis**, a process that uses the energy from the sun to convert carbon dioxide and water

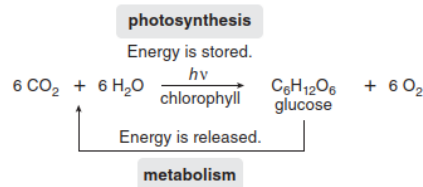
Figure 28.1
Some examples of
carbohydrates



These compounds illustrate the structural diversity of carbohydrates and their derivatives. **Glucose** is the most common simple sugar, whereas **cellulose**, which comprises wood, plant stems, and grass, is the most common carbohydrate in the plant world. **Doxorubicin**, an anticancer drug that has a carbohydrate ring as part of its structure, has been used in the treatment of leukemia, Hodgkin's disease, and cancers of the breast, bladder, and ovaries. **2'-Deoxyadenosine 5'-monophosphate** is one of the four nucleotides that form DNA.

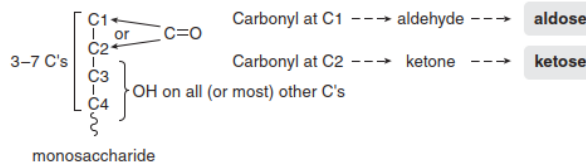
Although the metabolism of lipids provides more energy per gram than the metabolism of carbohydrates, glucose is the preferred source when a burst of energy is needed during exercise. Glucose is water soluble, so it can be quickly and easily transported through the bloodstream to the tissues.

into glucose and oxygen. This energy is released when glucose is metabolized. The oxidation of glucose is a multistep process that forms carbon dioxide, water, and a great deal of energy (Section 6.4).



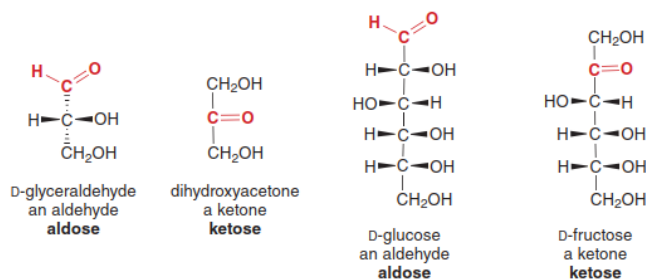
28.2 Monosaccharides

The simplest carbohydrates are called **monosaccharides** or **simple sugars**. **Monosaccharides have three to seven carbon atoms** in a chain, with a **carbonyl group** at either the terminal carbon (C1) or the carbon adjacent to it (C2). In most carbohydrates, each of the remaining carbon atoms has a **hydroxy group**. Monosaccharides are often drawn vertically, with the carbonyl group at the top. When this convention is used, monosaccharides look different from molecules encountered in prior chapters.



- Monosaccharides with an aldehyde carbonyl group at C1 are called aldoses.
- Monosaccharides with a ketone carbonyl group at C2 are called ketoses.

Several examples of simple carbohydrates are shown. D-Glyceraldehyde and dihydroxyacetone have the same molecular formula, so they are **constitutional isomers**, as are D-glucose and D-fructose.



All carbohydrates have common names. The simplest aldehyde, glyceraldehyde, and the simplest ketone, dihydroxyacetone, are the only monosaccharides whose names do not end in the suffix **-ose**. (The prefix “D-” is explained in Section 28.2C.)

A monosaccharide is called:

- a **triose** if it has 3 C's;
- a **tetrose** if it has 4 C's;
- a **pentose** if it has 5 C's;
- a **hexose** if it has 6 C's, and so forth.



D-Fructose is almost twice as sweet as normal table sugar (sucrose) with about the same number of calories per gram. “Lite” food products use only half as much fructose as sucrose for the same level of sweetness, and so they have fewer calories.



Dihydroxyacetone is the active ingredient in many artificial tanning agents.

25.4 Nucleic Acids

Nucleic acids are high molar mass polymers that play an essential role in protein synthesis. *Deoxyribonucleic acid (DNA)* and *ribonucleic acid (RNA)* are the two types of nucleic acid. DNA molecules are among the largest molecules known; they have molar masses of up to tens of billions of grams. On the other hand, RNA molecules vary greatly in size, some having a molar mass of about 25,000 g. Compared with proteins, which are made of up to 20 different amino acids, nucleic acids are fairly simple in composition. A DNA or RNA molecule contains only four types of building blocks: purines, pyrimidines, furanose sugars, and phosphate groups (Figure 25.17). Each purine or pyrimidine is called a *base*.

In the 1940s, Erwin Chargaff[†] studied DNA molecules obtained from various sources and observed certain regularities. *Chargaff's rules*, as his findings are now known, describe these patterns:

1. The amount of adenine (a purine) is equal to that of thymine (a pyrimidine); that is, $A = T$, or $A/T = 1$.
2. The amount of cytosine (a pyrimidine) is equal to that of guanine (a purine); that is, $C = G$, or $C/G = 1$.
3. The total number of purine bases is equal to the total number of pyrimidine bases; that is, $A + G = C + T$.

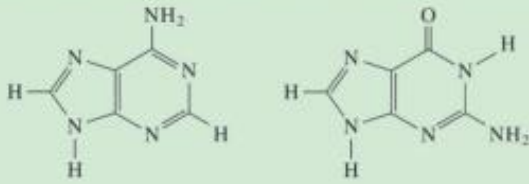
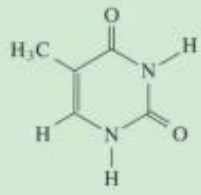
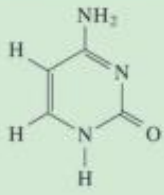
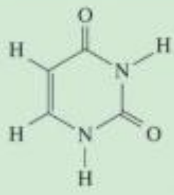
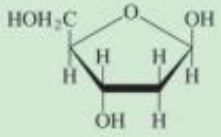
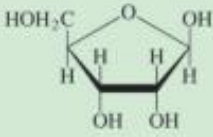
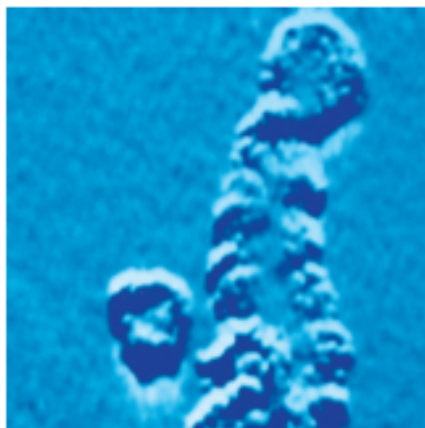
	Found only in DNA	Found in both DNA and RNA	Found only in RNA
Purines		 Adenine Guanine	
Pyrimidines	 Thymine	 Cytosine	 Uracil
Sugars	 Deoxyribose		 Ribose
Phosphate		 Phosphate	

Figure 25.17 *The components of the nucleic acids DNA and RNA.*



An electron micrograph of a DNA molecule. The double-helical structure is evident. If the DNA molecules from all the cells in a human were stretched and joined end to end, the length would be about 100 times the distance from Earth to the sun!

Source: Lawrence Berkeley National Laboratory

Based on chemical analyses and information obtained from X-ray diffraction measurements, James Watson[†] and Francis Crick[‡] formulated the double-helical structure for the DNA molecule in 1953. Watson and Crick determined that the DNA molecule has two helical strands. Each strand is made up of **nucleotides**, which consist of a base, a deoxyribose, and a phosphate group linked together (Figure 25.18).

The key to the double-helical structure of DNA is the formation of hydrogen bonds between bases in the two strands of a molecule. Although hydrogen bonds can form between any two bases, called *base pairs*, Watson and Crick found that the most favorable couplings are between adenine and thymine and between cytosine and guanine (Figure 25.19). Note that this scheme is consistent with Chargaff's rules, because every purine base is hydrogen-bonded to a pyrimidine base, and vice versa ($A + G = C + T$). Other attractive forces such as dipole-dipole

[†]James Dewey Watson (1928–). American biologist. Watson shared the 1962 Nobel Prize in Physiology or Medicine with Crick and Maurice Wilkins for their work on the DNA structure, which is considered by many to be the most significant development in biology in the twentieth century.

[‡]Francis Harry Compton Crick (1916–2004). British biologist. Crick started as a physicist but became interested in biology after reading the book *What Is Life?* by Erwin Schrödinger (see Chapter 7). In addition to elucidating the structure of DNA, for which he was a corecipient of the Nobel Prize in Physiology or Medicine in 1962, Crick made many significant contributions to molecular biology.

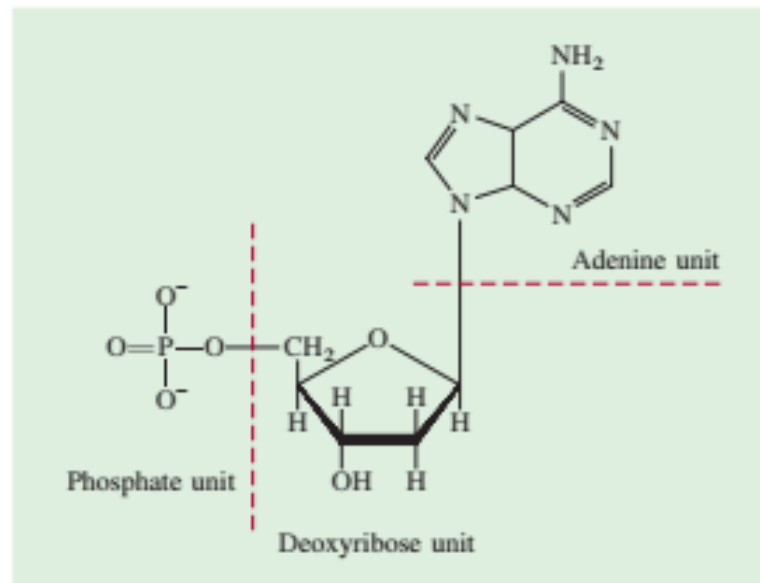


Figure 25.18 *Structure of a nucleotide, one of the repeating units in DNA.*

interactions and van der Waals forces between the base pairs also help to stabilize the double helix.

The structure of RNA differs from that of DNA in several respects. First, as shown in Figure 25.17, the four bases found in RNA molecules are adenine, cytosine, guanine, and uracil. Second, RNA contains the sugar ribose rather than the 2-deoxyribose of DNA. Third, chemical analysis shows that the composition of RNA does not obey Chargaff's rules. In other words, the purine-to-pyrimidine ratio is not equal to 1 as in the case of DNA. This and other evidence rule out a double-helical structure. In fact, the RNA molecule exists as a single-strand polynucleotide. There are actually three types

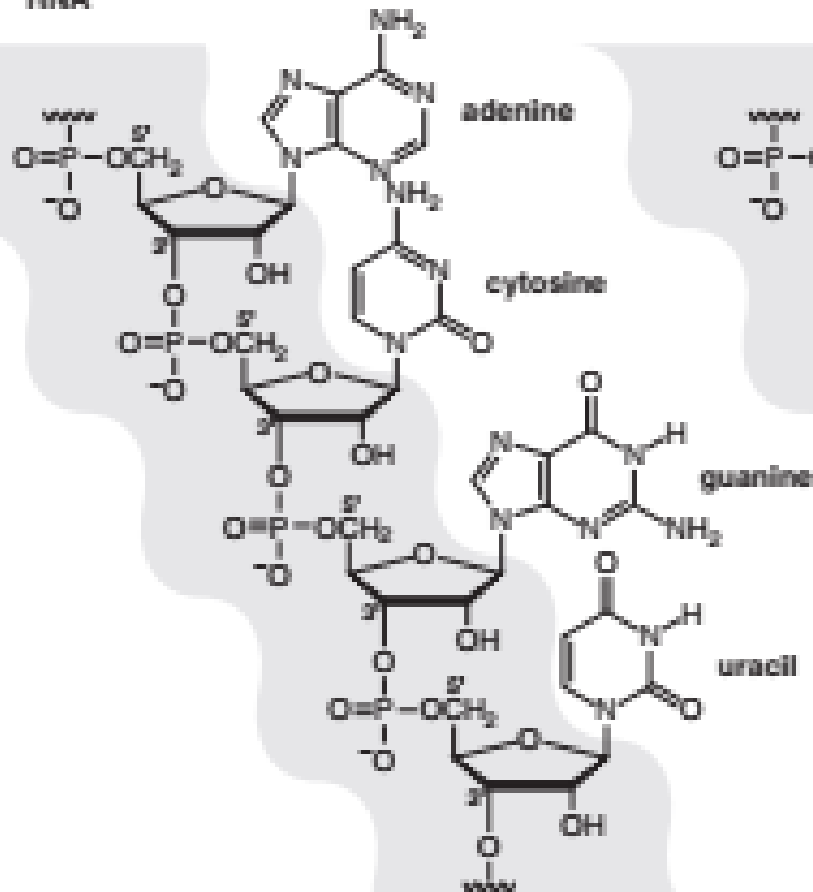
of RNA molecules—messenger RNA (*m*RNA), ribosomal RNA (*r*RNA), and transfer RNA (*t*RNA). These RNAs have similar nucleotides but differ from one another in molar mass, overall structure, and biological functions.

DNA and RNA molecules direct the synthesis of proteins in the cell, a subject that is beyond the scope of this book. Introductory texts in biochemistry and molecular biology explain this process.

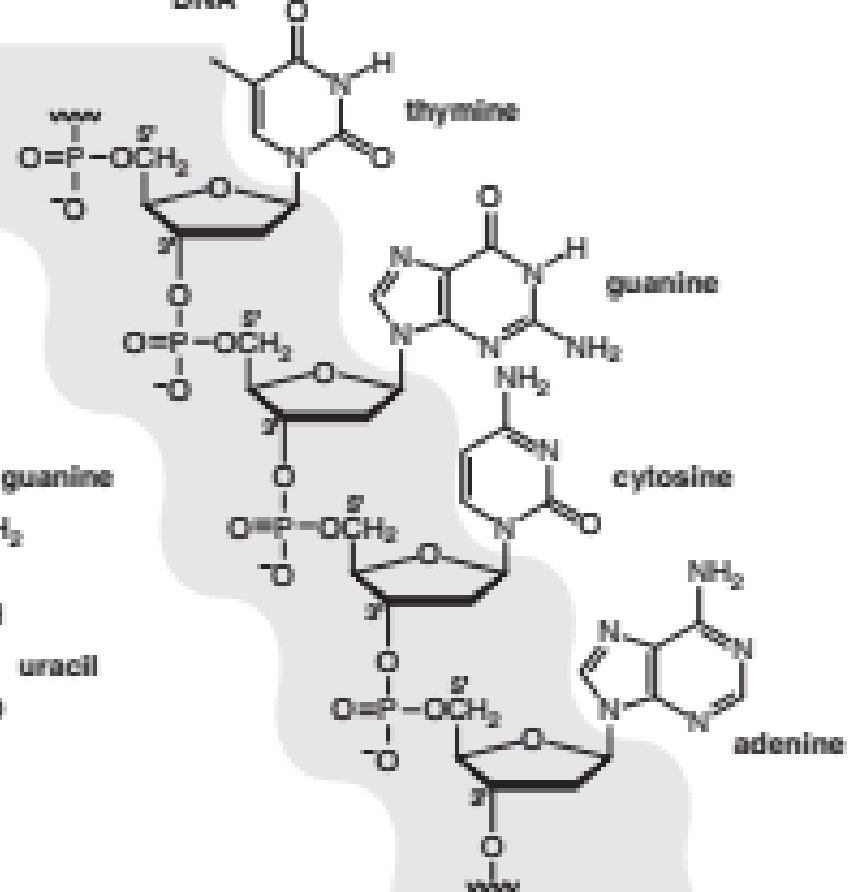
The Chemistry in Action essay “DNA Fingerprinting” describes a technique in crime investigation that is based on our knowledge of DNA sequence.

Figure 28.11 Short segments of RNA and DNA

**Ribonucleic acid
RNA**



**Deoxyribonucleic acid
DNA**



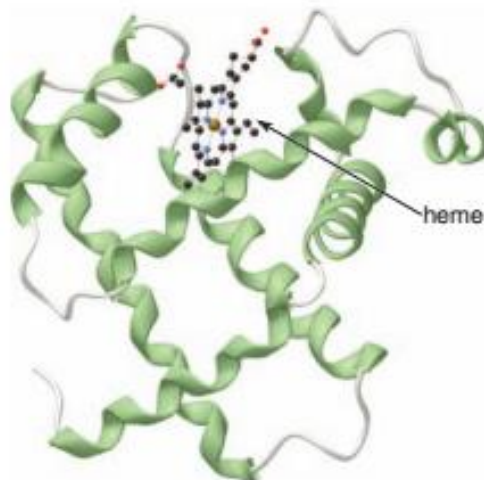
29.10C Hemoglobin and Myoglobin

Hemoglobin and **myoglobin**, two globular proteins, are called **conjugated proteins** because they are composed of a protein unit and a nonprotein molecule called a **prosthetic group**. The prosthetic group in hemoglobin and myoglobin is **heme**, a complex organic compound containing the

Figure 29.19

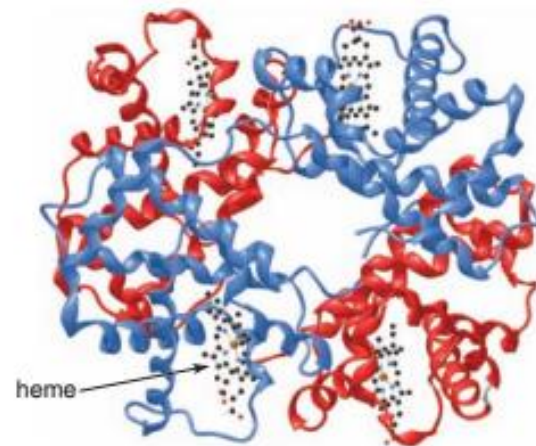
Protein ribbon diagrams for myoglobin and hemoglobin

a. Myoglobin



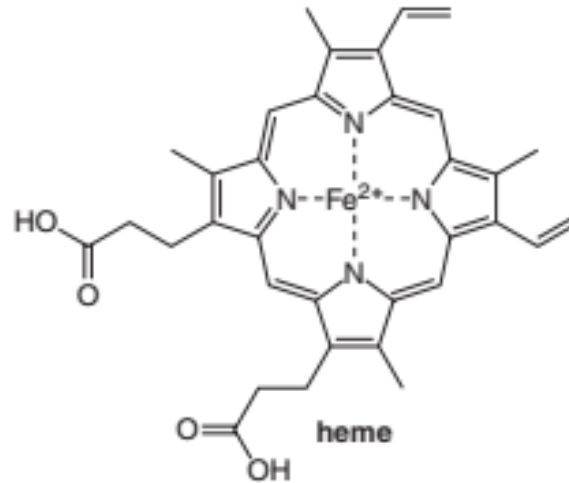
Myoglobin consists of a single polypeptide chain with a heme unit shown in a ball-and-stick model.

b. Hemoglobin



Hemoglobin consists of two α and two β chains shown in red and blue, respectively, and four heme units shown in ball-and-stick models.

Fe^{2+} ion complexed with a nitrogen heterocycle called a **porphyrin**. The Fe^{2+} ion of hemoglobin and myoglobin binds oxygen in the blood. Hemoglobin, which is present in red blood cells, transports oxygen to wherever it is needed in the body, whereas myoglobin stores oxygen in tissues. Ribbon diagrams for myoglobin and hemoglobin are shown in Figure 29.19.



Myoglobin, the chapter-opening molecule, has 153 amino acid residues in a single polypeptide chain. It has eight separate α -helical sections that fold back on one another, with the prosthetic heme group held in a cavity inside the polypeptide. Most of the polar residues are found on the outside of the protein so that they can interact with the water solvent. Spaces in the interior of the protein are filled with nonpolar amino acids. Myoglobin gives cardiac muscle its characteristic red color.

Hemoglobin consists of four polypeptide chains (two α subunits and two β subunits), each of which carries a heme unit. Hemoglobin has more nonpolar amino acids than myoglobin. When each subunit is folded, some of these remain on the surface. The van der Waals attraction between these hydrophobic groups is what stabilizes the quaternary structure of the four subunits.

6.11 Enzymes

The catalysts that synthesize and break down biomolecules in living organisms are governed by the same principles as the acids and metals in organic reactions. The catalysts in living organisms, however, are usually protein molecules called **enzymes**.

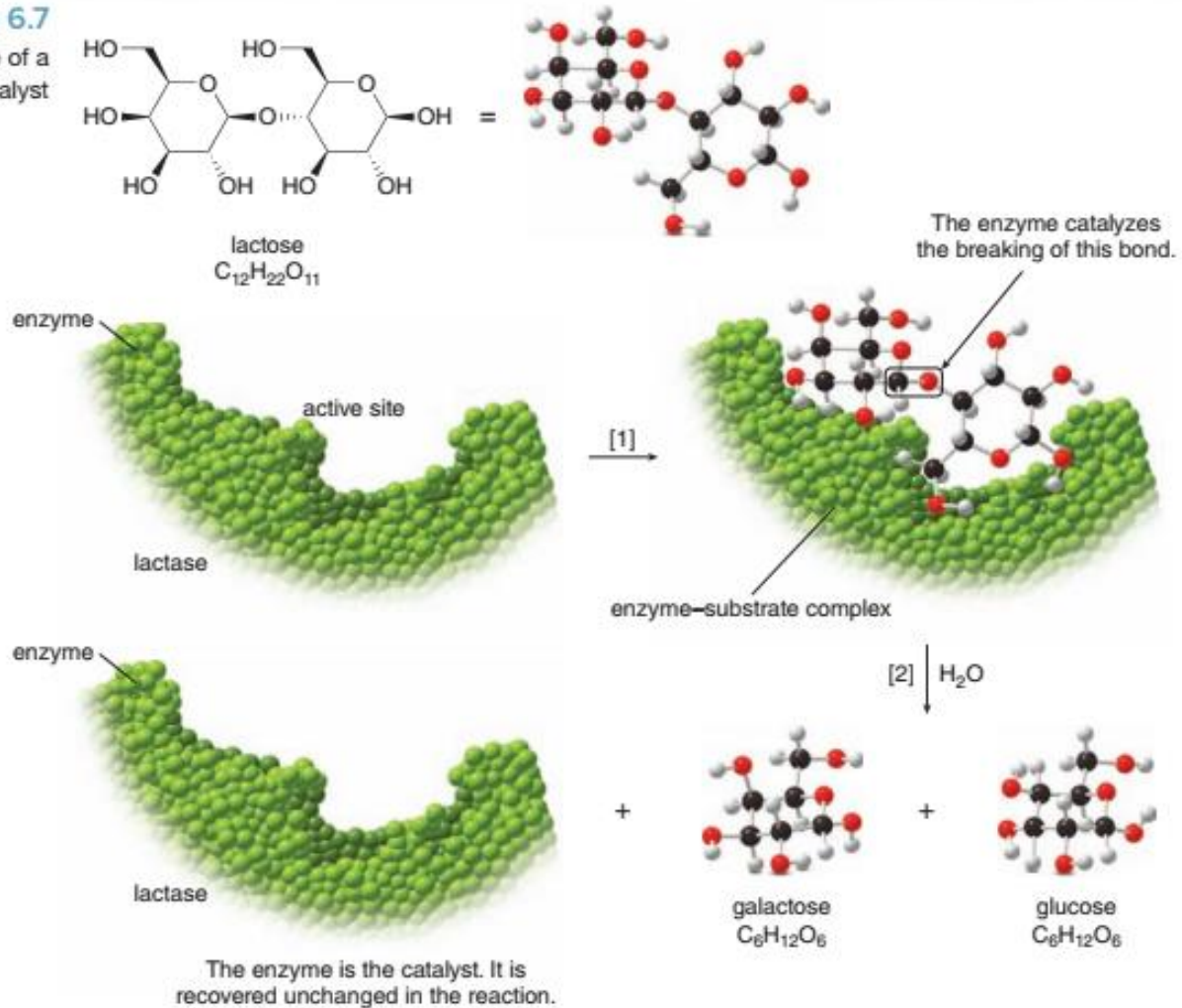
- *Enzymes* are biochemical catalysts composed of amino acids held together in a very specific three-dimensional shape.

An enzyme contains a region called its **active site**, which binds an organic reactant, called a **substrate**. When bound, this unit is called the **enzyme–substrate complex**, as shown schematically in Figure 6.7 for the enzyme lactase, the enzyme that binds lactose, the principal carbohydrate in milk. Once bound, the organic substrate undergoes a very specific reaction at an enhanced rate. In this example, lactose is converted into two simpler sugars, glucose and galactose. When individuals lack adequate amounts of lactase, they are unable to digest lactose, causing abdominal cramping and diarrhea.

An enzyme speeds up a biological reaction in a variety of ways. It may hold reactants in the proper conformation to facilitate reaction, or it may provide an acidic site needed for a particular transformation. Once the reaction is completed, the enzyme releases the substrate and it is then able to catalyze another reaction.

Figure 6.7

Lactase, an example of a biological catalyst



The enzyme lactase binds the carbohydrate lactose ($C_{12}H_{22}O_{11}$) in its active site in Step [1]. Lactose then reacts with water to break a bond and form two simpler sugars, galactose and glucose, in Step [2]. This process is the first step in digesting lactose, the principal carbohydrate in milk.